Results Report Phase T — Heuristics Miner (PM4Py)

Theoretical a-priori assessment from specification and documentation. Format follows the example reports; question wording at the level of the questionnaire update 2026-07-01.

Answers drawn from the characterization literature of the Heuristics Miner (Weijters et al. 2006; Weijters & Ribeiro 2011) and from the tooling documentation of the chosen setup (Berti et al. 2023, PM4Py).

Structured

Item	Answer	Rationale
T-S-BQ-1 Active restriction beyond the recorded traces, rather than mere log reproduction?	Yes	Dependency/frequency threshold actively restricts edges.
T-S-BQ-2 Documented guarantee that the observed variants remain replayable?	No	The Heuristics Miner filters frequency edges by default; no documented replay guarantee.
T-S-SF-1 Explicit, connected control-flow structure or losslessly equivalent notation?	Yes	Petri net with source/sink is directed and connected.

Semi-Structured

Item	Answer	Rationale
T-Sm-BQ-1 Can it enforce fixed local orderings on part of the behavior?	Yes	The Petri/heuristic net can firmly order local sequences, not merely apply a global restriction level.
T-Sm-BQ-2 Can it leave part of the behavior open/unordered?	Yes	AND-splits in the net allow concurrent, unordered parts.
T-Sm-SF-1 Fixed and open dependencies side by side in one model?	No	Single-mode imperative; fixed and open bindings are not represented side by side.

Loosely Structured

Item	Answer	Rationale
T-L-BQ-1 Evidence-driven individual decision per restriction (only with sufficient log support)?	Yes	Each dependency edge is retained individually via the dependency/frequency measure.
T-L-BQ-2 Documented generalization, so that the output accepts behavior beyond the exact traces?	Yes	The miner infers concurrency and abstracts via thresholds.
T-L-SF-1 Binding via local constraints rather than a fixed path structure?	No	Petri/C-net enforces a connected path structure.

Tfit Result

Structuredness class	#Yes / #Items	Tfit (%)
Structured	2/3	66.7
Semi-Structured	2/3	66.7

Structuredness class	#Yes / #Items	Tfit (%)
Loosely Structured	2/3	66.7

Interpretation. The a-priori profile is identical across all classes (66.7% / 66.7% / 66.7%). The documented mechanisms are consistently imperative: the miner restricts via a dependency/frequency threshold and reconstructs the dominant behavior into a connected Petri-net graph with start and end. This carries the active restriction (BQ-1), both reframed semi capabilities — firmly ordering local sequences (Sm-BQ-1) and leaving parts open/unordered (Sm-BQ-2) — and the evidence-driven, frequency-based generalization at the loose pole (L-BQ-1, L-BQ-2). Not carried are the documented replay guarantee (S-BQ-2) and the constraint-based notation that represents fixed and open bindings side by side (Sm-SF-1, L-SF-1). Note that this describes the a-priori capability; the empirical output (Phase E) tests what actually happens on the concrete logs.

Note on the setup. The empirical runs (Phase E) use the PM4Py default setup (`dependency_threshold 0.5`, `and_threshold 0.65`, `loop_two_threshold 0.5`, conformance `token_replay`). The a-priori assessment refers to the documented algorithm capability.

Sources. Weijters, A. J. M. M., van der Aalst, W. M. P. & Alves de Medeiros, A. K. (2006): Process Mining with the HeuristicsMiner Algorithm. · Weijters, A. J. M. M. & Ribeiro, J. T. S. (2011): Flexible Heuristics Miner (FHM). IEEE CIDM. · Berti, A., van Zelst, S. & Schuster, D. (2023): PM4Py, A Process Mining Library for Python.

Results Report — Phase E (empirical), Heuristics Miner (PM4Py) · all three structuredness classes · BQ & IN

1 Objective and procedure

The models discovered with the Heuristics Miner (PM4Py, `discover_petri_net_heuristics`, `dependency_threshold 0.5`) are evaluated against the empirical Phase E items of all three classes. Four items per class: two on Behavioral Quality (BQ), two on Interpretability (IN). Three-point scale per item (2 = full, 1 = partial, 0 = absent). Each item is averaged over the three logs of the class, then the class Efit is formed: Efit = sum of item means / 8 × 100.

Note on fitness. Unlike the Inductive Miner, the fitness of the Heuristics Miner is not guaranteed to be 1.0: the frequency filter can discard attested edges. On structured logs it stays at 1.000 (100% fitting traces); on semi- and loosely structured logs it drops (down to 0.759) and the fitting-traces rate declines markedly. Low fitness together with high precision is an overfitting signal and is read in the BQ assessment together with the model form.

2 Data basis: metrics of all models

Log	Cl.	Fitness	Fit-Traces	Precision	Silent	Edges	CFC	Simpl.
Log01_structured	S	1.000	100%	1.000	2	18	3	0.895
Log04_structured	S	1.000	100%	1.000	0	14	3	0.867
Log05_structured	S	1.000	100%	0.953	8	28	10	0.750
Log02_semiStructured	Sm	1.000	100%	0.907	15	42	17	0.615
Log06_semiStructured	Sm	0.969	25%	0.927	10	36	12	0.714
Log07_semiStructured	Sm	0.961	0%	0.814	9	38	13	0.689
Log03_looselyStructured	L	0.759	58%	0.898	3	12	5	0.714
Log08_looselyStructured	L	0.774	30%	0.943	4	18	6	0.714
Log11_looselyStructured	L	0.772	27%	0.975	13	44	17	0.571

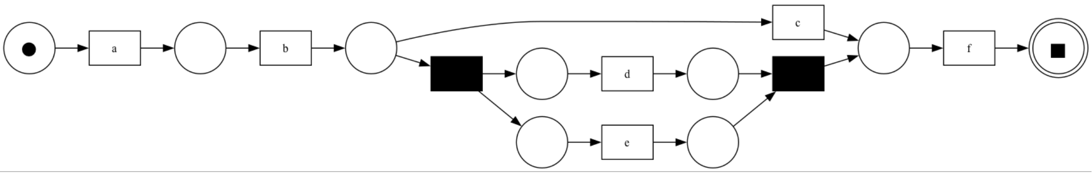
CFC = control-flow complexity, Simpl. = simplicity (arc-degree). Silent = invisible transitions from the heuristic-net-to-Petri-net mapping. The loosely runs were reproduced with the PM4Py default setup.

3 Class Structured — Efit = 91.7%

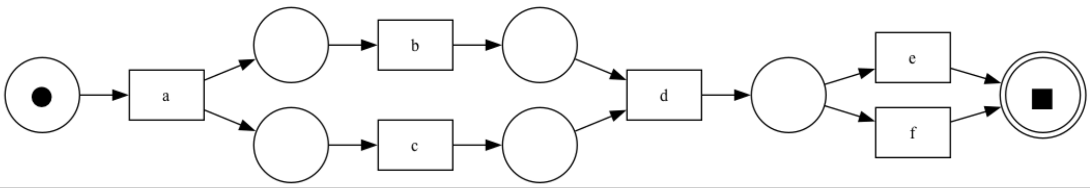
Logs: Log01, Log04, Log05. Behavioral Quality 100.0% · Interpretability 83.3%.

Item	Challenge	Log01	Log04	Log05	Mean	Question (short)
E-S-BQ-1	Behav. Quality	2	2	2	2.00	How tightly does the output bound the behavior space beyond the observed variants?
E-S-BQ-2	Behav. Quality	2	2	2	2.00	Do the variants attested in the log remain replayable?
E-S-IN-1	Interpret.	2	2	0	1.33	How compactly does the output represent the behavior?
E-S-IN-2	Interpret.	2	2	2	2.00	Can the execution ordering be read off from the output?
Efit sum					**7.33/8**	**= 91.7%**

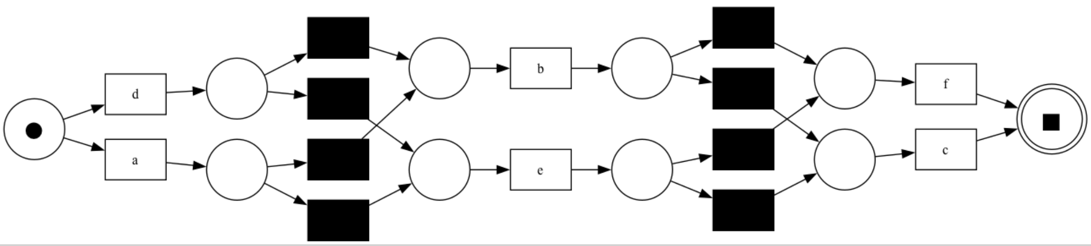
Rationale per item. E-S-BQ-1: precision 1.000 / 1.000 / 0.953, clear connected control flow without flower construct → all 2. E-S-BQ-2: fitness 1.000, 100% fitting traces, all attested variants replayable → all 2. E-S-IN-1: Log01 (CFC 3, Simpl 0.895) and Log04 (CFC 3, Simpl 0.867) are very compact → 2. Log05 has 8 silent transitions against 6 visible transitions, unnecessarily inflated → 0. E-S-IN-2: clear left-to-right flow with recognizable start/end → all 2.



Log01 · Structured · Petri net



Log04 · Structured · Petri net



Log05 · Structured · Petri net

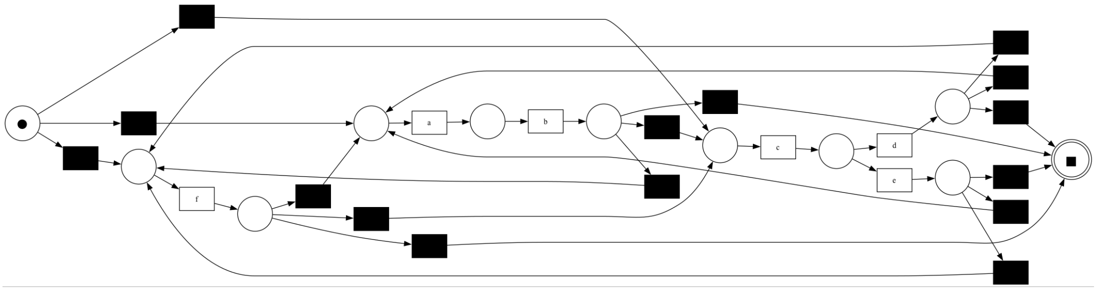
4 Class Semi-Structured — Efit = 70.8%

Logs: Log02, Log06, Log07. Behavioral Quality 83.3% · Interpretability 58.3%.

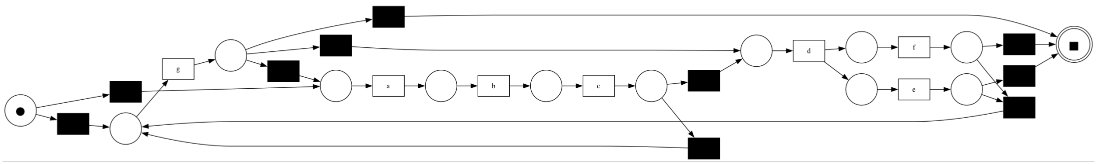
Item	Challenge	Log02	Log06	Log07	Mean	Question (short)
E-Sm-BQ-1	Behav. Quality	2	2	2	2.00	Does the model fix the internal ordering of the stable fragments?
E-Sm-BQ-2	Behav. Quality	2	1	1	1.33	Does the model allow fragment recombinations beyond the observed arrangement, with replay preserved?
E-Sm-IN-1	Interpret.	1	1	2	1.33	Are the stable and the flexible reference regions distinguishable?

Item	Challenge	Log02	Log06	Log07	Mean	Question (short)
E-Sm-IN-2	Interpret.	1	1	1	1.00	Is the output readable as one coherent model spanning both regions?
Efit sum					**5.67/8**	**= 70.8%**

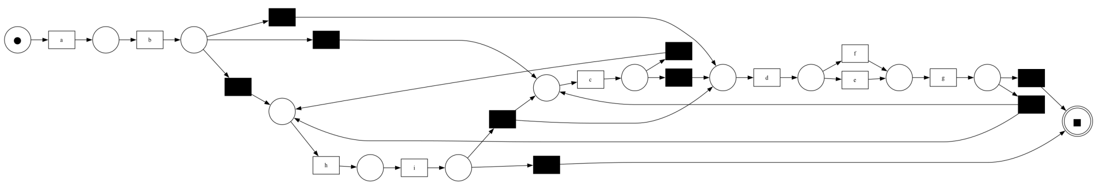
Rationale per item. E-Sm-BQ-1: the stable fragments keep their internal ordering (clear, firmly ordered control flow), while the flexible parts stay open (precision 0.81–0.93) → all 2. E-Sm-BQ-2: Log02 replays all variants (fitness 1.000) and additionally admits recombinations → 2. Log06 (fitness 0.969, 25% fitting traces): not all parts are connected, the fragment link d→f→abc is not visible → 1. Log07 (fitness 0.961, 0% fitting traces): consistent small partial failures → 1. E-Sm-IN-1: Log02 and Log06 are recognizable in blocks despite overload and crossing lines → 1. Log07 is somewhat more compactly readable → 2. E-Sm-IN-2: the spaghetti-like rendering reduces readability as one model spanning both regions → all 1.



Log02 · Semi-Structured · Petri net



Log06 · Semi-Structured · Petri net



Log07 · Semi-Structured · Petri net

5 Class Loosely Structured — Efit = 25.0%

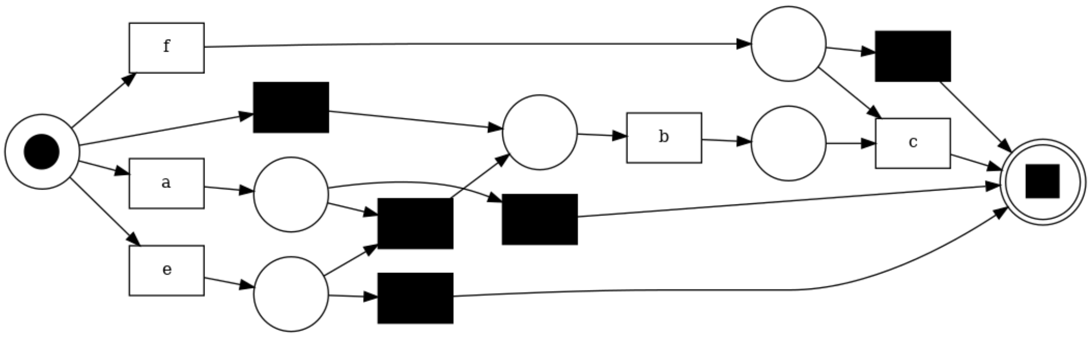
Logs: Log03, Log08, Log11. Behavioral Quality 16.7% · Interpretability 33.3%.

Note on the assessment. This class was previously ungraded; the models were reproduced (`dependency_threshold 0.5`, `token_replay`) and are shown below. The assessment was revised downward from a first, overly optimistic version and calibrated against the yardstick of the Inductive Miner (loosely Efit 29.2%): the reproduced nets show a fitness of only 0.71–0.82, i.e. 18–29% of the observed traces are not replayable. The high precision is therefore overfitting, not clean restriction, and does not translate into a BQ gain.

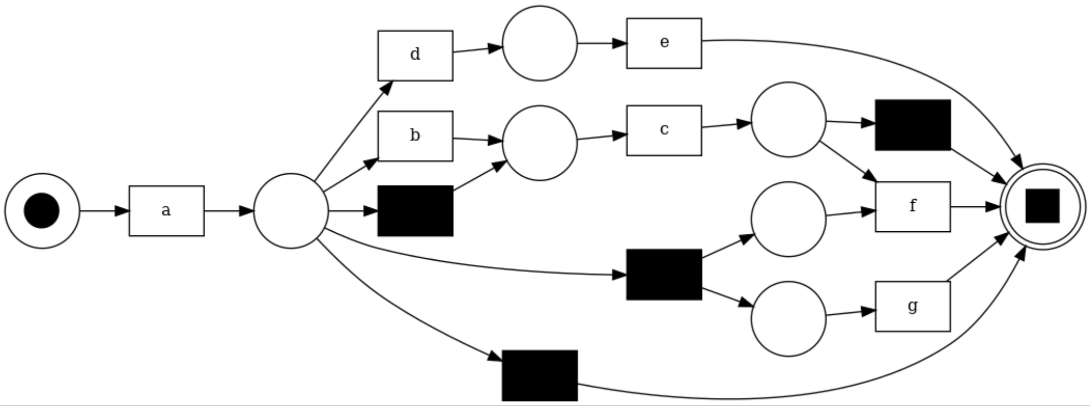
Item	Challenge	Log03	Log08	Log11	Mean	Question (short)
E-L-BQ-1	Behav. Quality	1	1	0	0.67	Does the model impose local restrictions that clearly exclude unsupported behavior?
E-L-BQ-2	Behav. Quality	0	0	0	0.00	Does the model allow behavior beyond the exactly observed variants, with replay preserved?
E-L-IN-1	Interpret.	1	1	0	0.67	Does the output condense to a few essential regularities?
E-L-IN-2	Interpret.	1	1	0	0.67	Does the output make local dependencies readable without forcing a global ordering?

Item	Challenge	Log03	Log08	Log11	Mean	Question (short)
Efit sum					**2.00/8**	**= 25.0%**

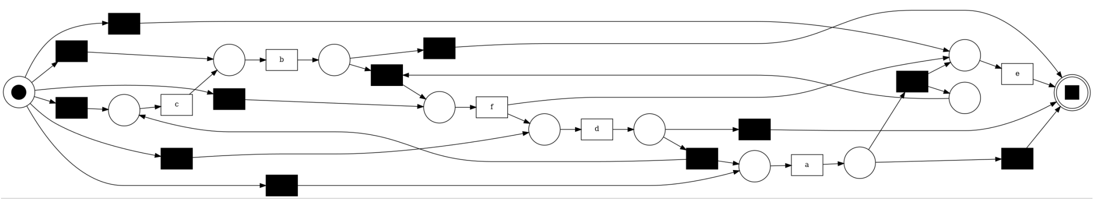
Rationale per item. E-L-BQ-1: the high precision (0.93–0.98) feigns restriction, but at fitness 0.71–0.82 it is overfitting — the net excludes attested behavior alongside the unsupported. Log03 and Log08 restrict recognizably locally, but overfitted → 1 each. Log11 (12 silent of 18 transitions, crossing edges, spaghetti) makes the restriction partly circumventable via τ → 0. E-L-BQ-2: uniformly 0. The new question requires generalization while preserving replay of the observed variants; exactly this fails: fitness is only 0.71–0.82, the models do not even fully replay the observed traces, and the high precision admits hardly any plausible additional behavior → all 0. This places the miner below the Inductive Miner on this item, which keeps the observed variants fully replayable. E-L-IN-1: Log03 (10 transitions, 5 silent) and Log08 (11 transitions, 4 silent) condense to a recognizable core but carry τ ballast → 1 each. Log11 (18 transitions, 12 silent, 40 edges) is a spaghetti net without condensation → 0. E-L-IN-2: Log03 and Log08 are small enough to read off local dependencies left to right; the imperative Petri-net notation nonetheless forces a global ordering → 1 each. Log11, with crossing edges, is not readable → 0.



Log03 · Loosely Structured · Petri net



Log08 · Loosely Structured · Petri net


Log11 · Loosely Structured · Petri net

6 Comparison of the classes

Class	BQ	IN	Efit
Structured	100.0%	83.3%	91.7%
Semi-Structured	83.3%	58.3%	70.8%
Loosely Structured	16.7%	33.3%	25.0%

BQ = mean of the two behavioral-quality items, normalized to the scale ceiling. IN = mean of the two interpretability items, normalized to the scale ceiling.

7 Classification

The empirical partial score falls monotonically with decreasing structuredness: 91.7% → 70.8% → 25.0%.

Structured (91.7%): structured logs are served almost perfectly. Deterministic behavior is rendered fully replayable and tightly bound (precision 0.95–1.00); only Log05 loses compactness due to eight silent transitions.

Semi-Structured (70.8%): here the miner's strength shows. On the behavioral side (83.3%) it fixes the internal ordering of the stable fragments and leaves the arrangement between them open. The deduction comes from interpretability (58.3%): the silent-transition mapping produces crossing lines that impair the separation of stable and flexible regions and the readability as one model.

Loosely Structured (25.0%): the miner's weakest class and clearly below the value initially recorded. The high precision (0.93–0.98) looks like restriction, but at fitness 0.71–0.82 it is overfitting: the models reconstruct only part of the exact variants and drop 18–29% of the observed traces. This collapses the behavioral side (BQ 16.7%) — BQ-2 falls to 0 because the replay of the observed variants required for loosely generalization fails, and BQ-1 is scored not as restriction but as overfitting. The miner thus falls below the Inductive Miner (29.2%), which at least keeps the observed variants fully replayable. The numerous silent transitions inflate the models (IN-1 low), and the imperative Petri-net rendering forces a global ordering (IN-2 low). Log11 tips into a spaghetti net with 12 silent transitions.